

HETEROKONTS

Heterokonts include some types of algae, which are photosynthesizing protoctists growing in or near water that do not have true leaves or roots.

Heterokonts are mostly defined as having two different types of flagella on the sperm they use in reproduction. One flagellum is covered in tiny, bristly hairs, or mastigonemes, and the other is smooth and whip-like. Heterokonts include diatoms and brown algae, as well as water moulds. Diatoms are single-celled algae with finely sculptured silica walls divided into two parts called valves. They are the dominant eukaryotic phytoplankton – plankton communities of small organisms that drift in open waters and engage in photosynthesis. Here, near the surface,

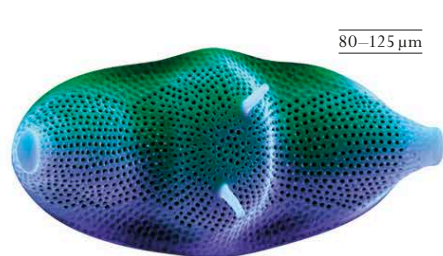
their pigments absorb energy from sunlight to make food. Diatoms contain green chlorophyll – as do plants – but also a brown pigment called fucoxanthin, which serves to broaden the spectrum of light that can be used, making their photosynthesis significantly more efficient.

Brown algae occupy coastal habitats around the world. They also make use of fucoxanthin, and have evolved into complex multicellular seaweeds that superficially resemble a plant. Rather than true roots and leaves, their body consists of a holdfast for clasping to rock and a creeping thallus that lacks the internal veins of a true leaf. Nevertheless, some brown seaweeds, such as kelps, can grow to extraordinary lengths and make extensive underwater forests on some offshore coastlines.

DOMAIN	EUKARYOTA
KINGDOM	PROTOCTISTA
CLADE	HETEROKONTOPHYTA
FAMILIES	177
SPECIES	About 20,000

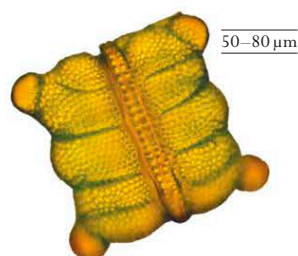
DEBATE FROM ALGA TO MOULD

Water moulds grow and feed like mould, but unlike true moulds, which are fungi, have plant-like cell walls and heterokont flagella, while some also cause plant diseases. DNA analysis links them with diatoms and brown algae, so perhaps they evolved from algae that abandoned their chloroplasts and turned parasitic.



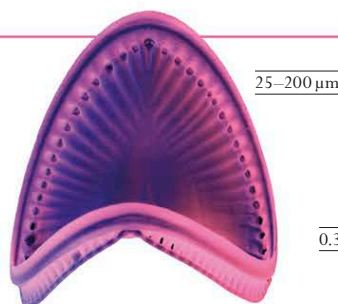
EPIPHYTIC DIATOM
Biddulphia sp.

The cause of brown film on fish tanks, in the wild this species grows on seaweed and rocks.



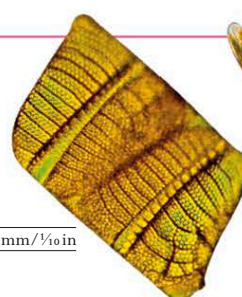
BARREL DIATOM
Biddulphia pulchella

This image clearly displays the different parts of a diatom: the two valves bound together by a narrow girdle.



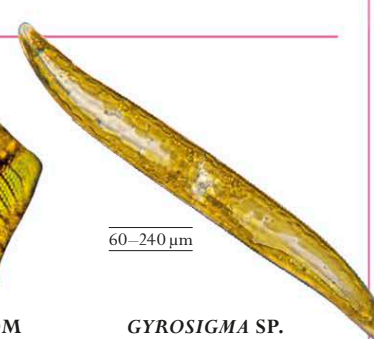
SADDLE DIATOM
Campylodiscus sp.

This genus of diatom shows a groove between the tubular lips, known as canals, around the edge of its valves.



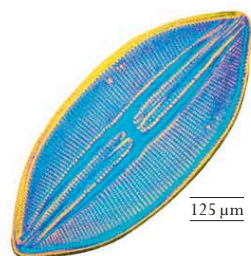
COLONIAL DIATOM
Isthmia nervosa

This species grows on other algae, specifically seaweeds, forming branching colonies.



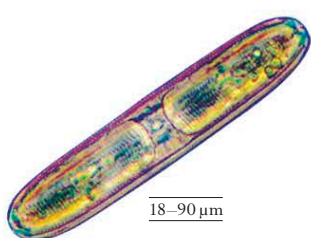
GYROSIGMA SP.

The name of this diatom refers to the sigmoidal curve of its cell, meaning that it follows a very slight S-shape.



SLIMY DIATOM
Lyrella lyra

This species secretes a gluey slime from its central groove to help the cells to glide along their host's surface.



POND DIATOM
Pinnularia sp.

Two chloroplasts are visible in this pen-shaped diatom, found in ponds and wet ground.

STEPHANODISCUS SP.

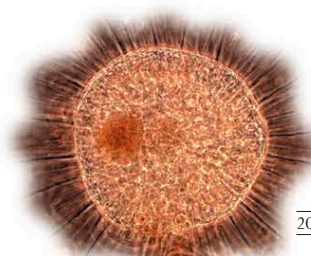
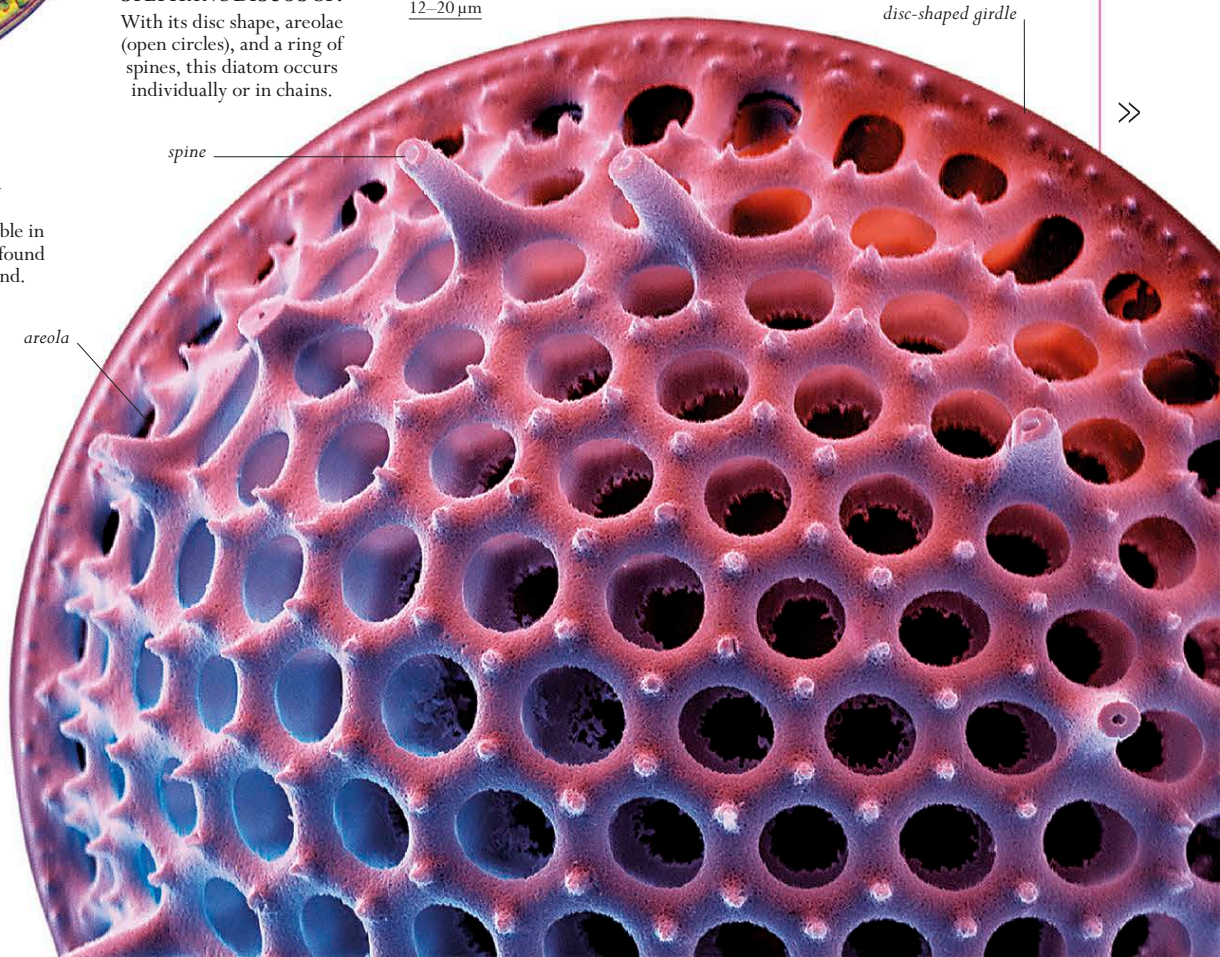
With its disc shape, areolae (open circles), and a ring of spines, this diatom occurs individually or in chains.

12–20 μm

spine

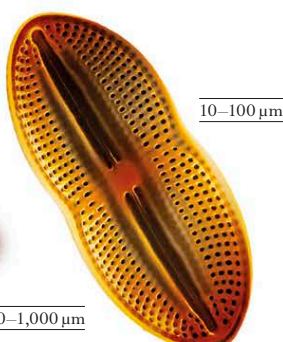
disc-shaped girdle

areola



URCHIN HELIOZOAN
Actinosphaerium sp.

This diatom resembles a sea urchin. It moves by shifting cell contents into its thin pseudopods.



GROOVED DIATOM
Diploneis sp.

The valves of this species of diatom appear as two heavily pronounced lips, known as canals, either side of a slit-like groove known as a raphe.